

Task 1.

dns

No.	Time	Source	Destination	Protocol	Length	Info
47	6.110184	10.45.171.156	10.45.171.242	DNS	79	Standard query 0xc529 HTTPS search-redirect.com
49	6.110459	10.45.171.156	10.45.171.242	DNS	79	Standard query 0x5960 AAAA search-redirect.com
50	6.110676	10.45.171.156	10.45.171.242	DNS	79	Standard query 0x11ab A search-redirect.com
51	6.110962	10.45.171.156	10.45.171.242	DNS	80	Standard query 0x892a HTTPS fonts.googleapis.com
52	6.111146	10.45.171.156	10.45.171.242	DNS	80	Standard query 0xe7c6 AAAA fonts.googleapis.com
53	6.111292	10.45.171.156	10.45.171.242	DNS	80	Standard query 0x93fa A fonts.googleapis.com
54	6.111497	10.45.171.156	10.45.171.242	DNS	80	Standard query 0x68f5 HTTPS cdnjs.cloudflare.com
55	6.111665	10.45.171.156	10.45.171.242	DNS	80	Standard query 0xeecf AAAA cdnjs.cloudflare.com
56	6.111839	10.45.171.156	10.45.171.242	DNS	80	Standard query 0x913a A cdnjs.cloudflare.com
57	6.112053	10.45.171.156	10.45.171.242	DNS	74	Standard query 0xefd5 HTTPS www.google.com
58	6.112232	10.45.171.156	10.45.171.242	DNS	74	Standard query 0x66dd AAAA www.google.com
59	6.112387	10.45.171.156	10.45.171.242	DNS	74	Standard query 0x222d A www.google.com
81	6.232750	10.45.171.242	10.45.171.156	DNS	157	Standard query response 0xc529 HTTPS search-redirect.com SOA ns-431.awsdns-53.com
82	6.233568	10.45.171.242	10.45.171.156	DNS	157	Standard query response 0x5960 AAAA search-redirect.com SOA ns-431.awsdns-53.com
83	6.234700	10.45.171.242	10.45.171.156	DNS	280	Standard query response 0x11ab A search-redirect.com A 18.161.97.93 A 18.161.97.4 A 18.161.97.69 A 18.161.97.31 NS ns-1191.awsdns-53.com
84	6.235014	10.45.171.242	10.45.171.156	DNS	137	Standard query response 0x892a HTTPS fonts.googleapis.com SOA ns1.google.com
85	6.235599	10.45.171.156	10.45.171.242	DNS	87	Standard query 0x5ee3 HTTPS content.search-redirect.com

Frame 47: 79 bytes on wire (632 bits), 79 bytes captured (632 bits) on interface \Device\NPF_{9BCFE89B-7000-4e15-ca08-6d1060ff9e4f} ca 82 08 00 45 00 N...m...0...E...

Ethernet II, Src: AzureWaveTec_4f:ce:82 (60:ff:9e:4f:ce:82), Dst: 4e:15:ca:08:6d:10 (4e:15:ca:08:6d:10)
Internet Protocol Version 4, Src: 10.45.171.156, Dst: 10.45.171.242
User Datagram Protocol, Src Port: 59371, Dst Port: 53
Domain Name System (query)

0000 4e 15 ca 08 6d 10 60 ff 9e 4f ce 82 08 00 45 00 N...m...0...E...
0010 00 41 06 e9 00 00 80 11 c7 da 8a 2d ab 9c 0a 2d A...e...e...
0020 ab f2 e7 eb 00 35 00 2d 34 81 c5 29 01 00 00 015...4...
0030 00 00 00 00 00 00 0f 73 65 61 72 63 68 2d 72 65s earch-re
0040 64 69 72 65 63 74 03 63 6f 6d 00 00 41 00 01 direct.c om=A...

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No.	Time	Source	Destination	Protocol	Length	Info
22	3.326257	2401:4900:54c0:2336...	2404:6800:4009:832...	QUIC	1292	Initial, DCID=81f8c2b9502cb6d0, PKN: 3, CRYPTO, PADDING, CRYPTO, PING, CRYPTO, CRYPTO, PADDING, PING, CRYPTO, CRYPTO, PADDING, PING
29	3.385619	2404:6800:4009:832...	2401:4900:54c0:2336...	QUIC	1292	Initial, SCID=e1f8c2b9502cb6d0, PKN: 5, CRYPTO, PADDING
45	6.028475	2401:4900:54c0:2336...	2606:4700:90d1:d8b7...	TLSv1.2	164	Application Data
46	6.099549	2401:4900:54c0:2336...	2606:4700:90d1:d8b7...	TLSv1.2	164	Application Data
60	6.114548	10.45.171.156	18.161.97.93	TLSv1.2	145	Application Data
61	6.114643	10.45.171.156	18.161.97.93	TLSv1.2	93	Application Data
69	6.115405	2401:4900:54c0:2336...	2606:4700:90d2:7cbc...	TLSv1.2	595	Application Data
70	6.115601	2401:4900:54c0:2336...	2606:4700:90d2:7cbc...	TLSv1.2	113	Application Data
72	6.115720	2401:4900:54c0:2336...	2606:4700:90d2:7cbc...	TLSv1.2	584	Application Data
80	6.187414	2606:4700:90d2:7cbc...	2401:4900:54c0:2336...	TLSv1.2	113	Application Data
95	6.237745	2401:4900:54c0:2336...	2404:6800:4000:1025...	QUIC	1292	Initial, DCID=b88271ed885fd970, PKN: 3, PADDING, PING, PADDING, CRYPTO, CRYPTO, PING, PADDING, CRYPTO, PADDING, CRYPTO, CRYPTO, PING
101	6.242198	2606:4700:90d1:d8b7...	2401:4900:54c0:2336...	TLSv1.2	153	Application Data
103	6.243941	2401:4900:54c0:2336...	2606:4700:90d2:e632...	QUIC	1292	Initial, DCID=676f865298baca71, PKN: 2, PING, PING, PADDING, CRYPTO, CRYPTO, PADDING, PING, CRYPTO, CRYPTO, PING, PING, PING, PING
109	6.267332	18.161.97.93	10.45.171.156	TLSv1.2	93	Application Data
121	6.286393	2404:6800:4000:1025...	2401:4900:54c0:2336...	QUIC	1292	Initial, SCID=f88271ed885fd970, PKN: 5, CRYPTO, PADDING
129	6.306240	2606:4700:90d1:d8b7...	2401:4900:54c0:2336...	TLSv1.2	153	Application Data
133	6.320958	2606:4700:90d2:e632...	2401:4900:54c0:2336...	QUIC	1262	Initial, SCID=01a860b85f8ca5219caa9db8f48caf049156dc8, PKN: 1. ACK. CRYPTO

> Frame 46: 164 bytes on wire (1312 bits), 164 bytes captured (1312 bits) on interface Vde:\Device\NPF_{9BCFEF...}

> Ethernet II, Src: AzureWaveTec_Afc:ce:82 (60:ff:9e:4f:ce:82), Dst: 4e:15:ca:08:6d:10 (4e:15:ca:08:6d:10)

> Internet Protocol Version 6, Src: 2401:4900:54c0:2336:cfff:395:d9fe:f135, Dst: 2606:4700:90d1:d8b7:9a38

> Transmission Control Protocol, Src Port: 60244, Dst Port: 443, Seq: 91, Ack: 1, Len: 90

> Transport Layer Security

```

0000  4e 15 ca 08 6d 10 60 ff  9e 4f ce 82 86 dd 60 08  N...m...O...
0010  53 2a 00 06 0e fe 24 01  49 00 54 c0 23 36 fc ff  S*.n...I.T#6...
0020  03 95 d9 fe f1 35 26 06  47 00 90 d1 d8 b7 9a 38  ....S8.G...-8
0030  04 e4 cc fs ac 50 eb 54  01 bb 12 36 b4 6f fe 56  ....P.T...6-o
0040  fd 1d 50 18 00 ff 5b b9  00 10 17 03 03 00 55 2d  ...P...[...U-
0050  b2 e2 cf b6 26 27 4d 6c  7e df db 47 75 06 dd 8b  ...&*ML...Ou-m
0060  7d 47 64 20 5c 33 a8 ba  90 15 7a 39 6a a8 b9 09  ...d B...25j...
0070  97 52 a7 60 80 27 2d b6  19 99 23 80 61 e0 36 c4  R...-...# a 6-
0080  c7 9e 64 53 72 8b ff df  aa d3 2a ef dc db 5d a9  ...jS...[*...]-
0090  95 39 a5 c9 fa 20 99 da  f2 93 e5 7a 36 39 7f bc  .9...-...z69...
00a0  50 c3 30 4e

```

Packet No.	Protocol	Info
4496	DNS	Standard query 0x1ef0 AAAA www.spotify.com
4914	DNS	Standard query 0xb5d4 A www.google.co.in
45	TLS (HTTPS)	Application Data
1262	TLS (HTTPS)	Client Hello (SNI=open.spotify.com)

Task 2.

Simulate a DNS lookup for `www.flipkart.com` using the `nslookup` command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

```
C:\Users\sketan>nslookup www.flipkart.com
Server: UnKnown
Address: 10.45.171.242

Non-authoritative answer:
Name: flipkart.com
Address: 163.53.76.86
Aliases: www.flipkart.com

C:\Users\sketan>
```

The IP address returned by `nslookup` is the numerical address of the server hosting the Flipkart website. Your browser uses this IP address to establish a connection to the website after converting the domain name into an IP address.

Task 3.

Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

HTTP	HTTPS
Data is not encrypted.	Data is encrypted using TLS/SSL.
Uses Port 80.	Uses Port 443.
Less secure and vulnerable to attacks.	More secure and protects user information.

Real-World Example

Google (<https://www.google.com>) uses HTTPS to securely transmit search queries and account information between the browser and Google's servers.

Task 4.

Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.

Packet No.	Protocol	Info
2353	DNS	Standard query 0x47a6 A www.ilovepdf.com
2461	DNS	Standard query response 0x1558 AAAA www.ilovepdf.com AAAA 2606:4700:90d3:5245:e9a5:4e4:4c2f:d469
2355	DNS	Standard query 0x1558 AAAA www.ilovepdf.com
2461	DNS	Standard query response 0x1558 AAAA www.ilovepdf.com AAAA 2606:4700:90d3:5245:e9a5:4e4:4c2f:d469

Observation

When the browser requests a new website, it first sends a Standard query asking for the IP address. The DNS server then replies with a Standard query response containing the IPv4 (A record) and sometimes IPv6 (AAAA record) addresses. The browser then uses these addresses to connect to the website.

Task 5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

DHCP (Dynamic Host Configuration Protocol)

- When you visit a cafe and connect your phone to the free Wi-Fi before opening Instagram.
- The DHCP server automatically assigns your phone an IP address, subnet mask, default gateway, and DNS server.
- Without DHCP, you would have to enter these network settings manually.
- Once the IP address is assigned, your phone can access the internet and load Instagram normally.